

Binding energy model for CRISPR-Cas systems

BIM254 Course project

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1 Introduction

Bacteria employ an adaptive immune systems to protect themselves against foreign invaders such as viruses. CRISPR or Clustered Regularly Interspaced Short Palindromic Repeats are small segments of the viral DNA captured by bacteria. In the event of another attack by the same virus, the stored segments in the form of RNAs with the help of an enzyme(Cas-CRISPR associated protein) identify the sequence specific to that virus and cuts the viral DNA, rendering it inactive (Nuñez et al.,2015). CRISPR-Cas sytems are classified into 2 classes, 6 types and several subtypes. Class I systems have multiple Cas protein, whereas class II have a single, large Cas protein. The CRISPR-Cas system is useful in genome editing applications and Cas9, Cas12a, and Cas13 have found success in the past decade. While Cas13 modifies RNA, Cas9 and Cas12a, which are used in this study, modify dsDNA(Koonin and Makarova,2019).

1.1 Mechanism

There are two main actors in the CRISPR tool: guide RNA and Cas protein. These two actors form the riboprotein complex(RNP) which binds to the specific DNA location.Two units typically make up guide RNA: a ~ 20 nt long sequence that complements the target region and a ~ 80 nt long sequence that is required for stability and Cas protein interaction. The former region is CRISPR RNA or crRNA, while the latter is the transactivating RNA or trRNA. The trRNA region being the most stable region would consist of hairpin loops and other motifs, while crRNA will be in its unfolded state before binding. In this study, we would be referring to crRNA as guide RNA(Koonin and Makarova,2019).

Cas protein recognizes a unique sequence of 2-4 nucleotides called PAM (Protospacer Adjacent Motif) in the target DNA and then binds to it. Once binded, guide RNA starts its invasion into the DNA duplex and must displace the strand adjacent to PAM. RNA-DNA hydribidization here is a competition between formation of hydrogen bonds between RNA-DNA and breaking of hydrogen bonds in the double stranded DNA resulting in the formation of an R-loop structure. Further, a stable binding leads to the cleavage of the target DNA sequence by the

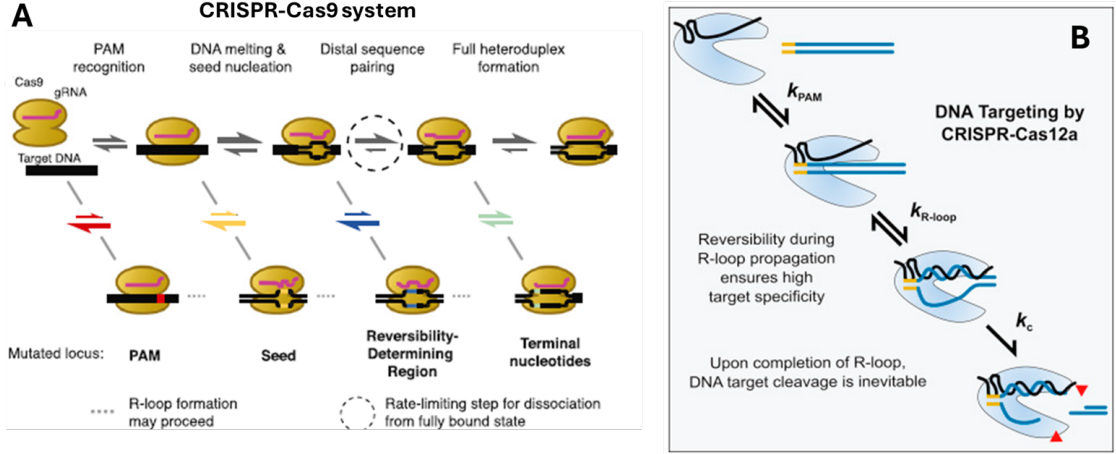


Figure 1: Binding unbinding mechanism of CRISPR-Cas9 system with a rate-limiting R-loop dissociation(Boyle et al.,2017), Cas12a DNA binding/unbinding with readily reversible R-loop formation(Strohkendl et al.,2018)

Cas protein(Xu et al.,2017). PAM for Cas9: NGG (most common and canonical), NAG and NGA(non-canonical) where N=any nucleotide. There are other PAM mutants, but the sequences with non-canonical NAG/NGA were more-bound to Cas9 in comparison (Boyle et al.,2017). The strand will look like this 5' end- 20 nt protospacer sequence – PAM -3' end. PAM for Cas12a: TTTN (most common) where N = any nucleotide(Strohkendl et al.,2018).The strand will be 5' end- PAM - 20 nt protospacer sequence -3' end.

Eventhough it is the dsDNA that is edited in both Cas9 and Cas12a systems, positioning of PAM sequence with respect to the protospacer, mechanism of RNA-DNA binding, the way in which cleavage occurs is different in both cases.

1.2 Binding/unbinding of Cas9 systems

The kinetic process of binding and unbinding for a CRISPR-Cas9 system is shown in Figure 1A. The first few nucleotides in the PAM-proximal regions (nucleotides 1–7) attach to the RNA after Cas9 binds to the PAM sequence. Mismatches in this region would lead to rapid dissociation in the PAM-Cas9 binding. Binding in the PAM-distal region (nucleotides 8–17) comes next. After stability is established, the RNP complex's rate-limiting unbinding takes longer. The stability is further increased by binding at the 18th to 20th sites. There are two nuclease domains

in Cas9, HNH and RuvC. HNH cuts the strand bound to the RNA, while NuVC cuts the other strand that is next to PAM sequence. Cas9 undergoes a change in conformation after ensuring stability activates both the nuclease domains which cuts DNA at both ends(Boyle et al.,2017).

1.3 Binding/unbinding of Cas12a systems

Cas12a/RNA binds to its PAM sequence TTTN and RNA begins pairing with its complementary strand in the DNA. If the length of base pairing is 17 and there are mismatches after that, the RNP can easily dissociate from the DNA. If there are no or few mismatches in the terminal region, the single RuVC nuclease domain makes cut in the region binded to gRNA. Then Cas12a undergoes a conformational change, positioning the RuVC domain to cut a different site from the other strand. Cas9 and Cas12a differ in that the former may dissociate following R-loop formation extremely slowly, whereas the latter dissociates quickly. Also, stability of RNA-DNA complex depends on binding throughout the entire 20-nt region and does not make the seed region special(Strohkendl et al.,2018). Strong binding in seed region is essential in Cas9 because Cas9 has to disrupt atleast two G-C bonds, whereas Cas12a has to break two A-T bonds.

1.4 Off-targeting

Cleaving the target sequence requires stable binding. However, sometimes the RNA binds to unintended genome and cuts that part. This could happen if researchers target a common sequence in a gene family rather than tailoring the guide RNA sequence for a specific DNA target. Such off-targeting can have negative impacts on human health. For example, if the guide RNA binds with a gene that suppresses cancer, there are risks involved that makes the cell cancerous.

Such off-targets depend mostly on the gRNA and those are mostly homologous to the target region. Fewer mismatches would mean the binding is more stable resulting in cleavage and it is reported that off-targets can have up to six mismatches. Experimental information on the type and position of mismatches has led to development of various off target scoring methods. Alkan et al. (2018) developed a biophysical model for off-target scores by making use of free energy parameters of

duplexes.

2 Binding energy model

The free energy of binding is approximated using four terms:

1. Free energy of RNA-DNA hybridization, ΔG_H given by

$$\Delta G_H[g, t] = \sum_{i=1}^{19} \Gamma_{\text{Cas9}}[i] \cdot \Delta G_{\text{RNA:DNA}}^{g[i,i+1]:t[i,i+1]} \quad (1 \leq i \leq 19) \quad (1)$$

where $\Gamma_{\text{Cas9}}[i]$ is the weight depending on the position of binding and $\Delta G_{\text{RNA:DNA}}^{g[i,i+1]:t[i,i+1]}$ is the energy contribution of the match between RNA and DNA at position i . The RNA-DNA parameters are estimated using Sugimoto et al.(1995) and Watkins et al.(2011) energy model.

2. RNA unfolding energy, ΔG_U , ViennaRNA -d2 model(Hofacker,2003) is used to calculate this energy for 20 nt gRNA sequence and Turner et al.(2010) model is used.
3. Energy contributions for opening the DNA, ΔG_O

$$\Delta G_O[t] = \sum_{i=1}^{19} \Delta G_{\text{DNA:DNA}}^{t'[i,i+1]:t[i,i+1]} \quad (2)$$

SantaLucia et al.(2004) nearest neighbor energy model is used here.

4. Final correction depending on the PAM sequence, δ_{PAM} . $\delta_{\text{PAM}} = 1$ for NGG and TTTN, $\delta_{\text{PAM}} = 0.9$ for NAG and $\delta_{\text{PAM}} = 0.8$ for NGA. Most of the sequences used in this study contains canonical PAM sequences and hence PAM correction is not used for most cases.

The approximate free energy of binding is given by:

$$\Delta G_B[g, t] = \delta_{\text{PAM}} \cdot (G_H[g, t] - \Delta G_O[t] - \Delta G_U[g])$$

(3)

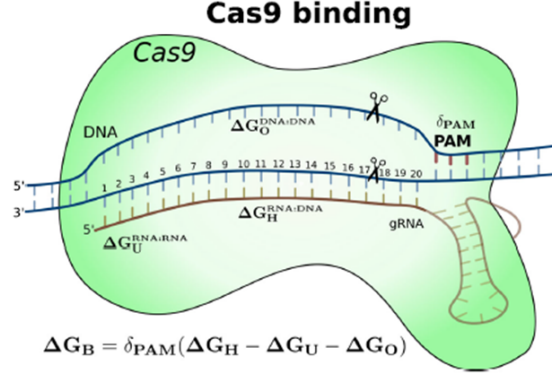


Figure 2: Binding energy model for RNA-DNA interaction(Alkan et al.,2018)

2.1 Binding stability

The weight is given by

$$w_i = \sum_{n=1}^{|O|} k_n \times \Delta G_{\tilde{g}[i,i+1]:\tilde{o}_n[i,i+1]}^{RNA:DNA} \quad (1 \leq i \leq 19)$$

(4)

For a particular gRNA, $\tilde{g}$, energy contributions are computed for all 19 base pair stacks between that gRNA and each off-target, $\tilde{o}_n$ with association rate constant k_n already known. The positional weights obtained from the summation is normalized to give the binding stability, $\Gamma_{Cas9} = [1.80, 1.96, 1.90, 2.13, 1.38, 1.46, 1.00, 1.39, 1.51, 1.98, 1.88, 1.72, 2.02, 1.93, 2.08, 1.94, 2.15, 2.04, 2.25]$. Consistently higher values in the PAM-proximal region demonstrate its significance in Cas9 system(Alkan et al.,2018).

3 Association rate constant

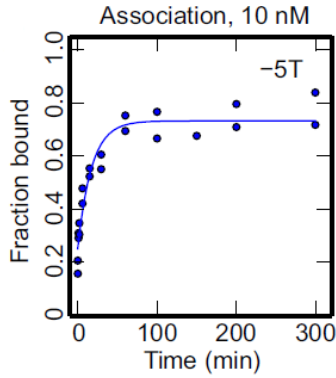
Alkan et al. (2018) used association rate of nuclease-dead Cas9 interactions with trace amounts of dsDNA as measured by Boyle et al. (2017). Fraction of Cas9-gRNA bound, f is determined by fluorescence in real time and it increases exponentially before getting saturated as shown in Figure 3A. Here, Cas9 concentration of 10 nM is used and there is a "T" mismatch at 5th nucleotide from the PAM sequence. This

association curve is fitted to obtain observed rate constant, k_{obs} with units of $time^{-1}$.

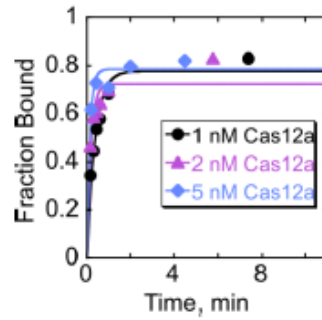
$$f = (f_{\max} - f_{\min})(1 - \exp(-k_{obs} \cdot t)) + f_{\min} \quad (5)$$

For each off-target sequence, k_{obs} was obtained for different concentrations. The slope of k_{obs} vs $[Cas9]_{initial}$ gives the association rate constant, k_{bind} for that off-target. $k_{obs} = k_{bind}[Cas9]_{initial} + k_{unbind}$.

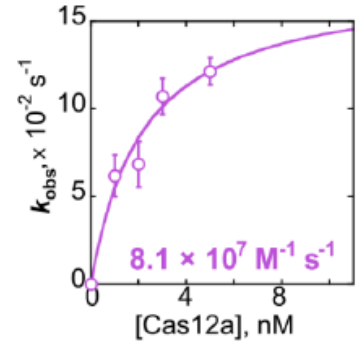
84,554 sequences were used for the binding experiments which included all possible single and double mismatches and close to 70,000 multiple mismatch cases(Boyle et al.,2017). The initial association rates of dCas9 were very reproducible for both single and double mismatches($R^2 = 0.96$), which reduced slightly with more mismatches ($R^2 = 0.86$).



(a) Fraction of 10nM Cas9/gRNA bound to off-target sequence with a mismatch "T" at the 5th nucleotide from PAM(Boyle et al.,2017)



(b) Fraction of Cas12a/gRNA bound to some off-target sequence(Strohkendl et al.,2018)



(c) Fitting of observed rate constant and Cas12a cocentration to obtain association rate constant(Strohkendl et al.,2018)

3.1 Association rate constant with a smaller subset

A smaller subset of off-target data was used to obtain the binding stability and these values are compared to the set of values used in Alkan et al. (2018). This is to verify if this is applicable for all Cas9 off-targets. For this purpose, association rate constant for active nuclease Cas9 from Pacesa et al. (2022) is used. The dataset consists of 4 different guide RNAs with a total of 16 off-target sequences.

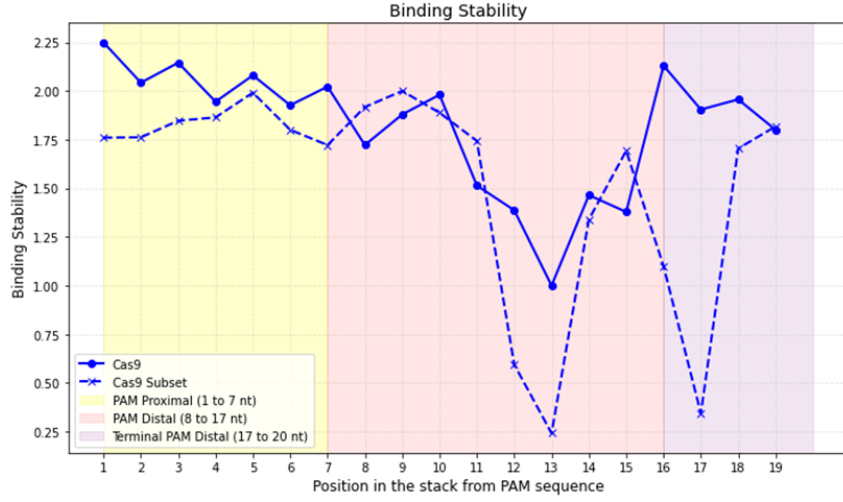


Figure 4: Binding stability for Cas9 system with overall set of 80000+ off-target sequences compared with a smaller subset of 16 off-targets

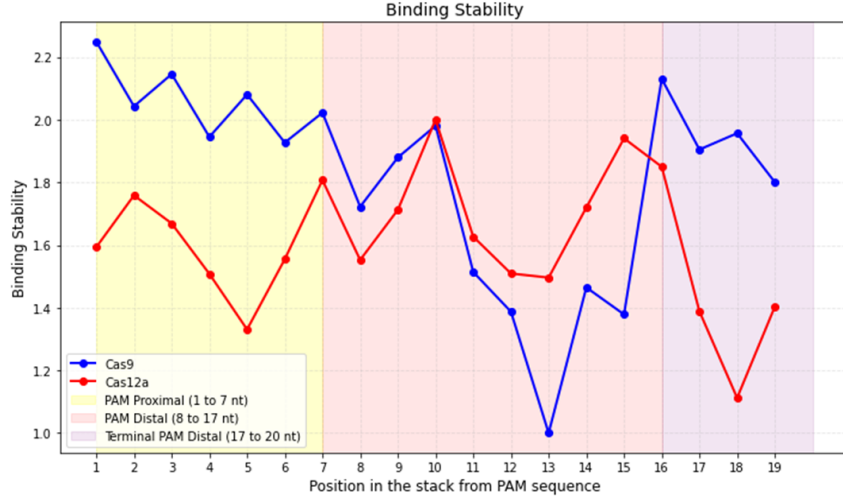


Figure 5: Binding stability for Cas9 and Cas12a

3.2 Association rate constant for Cas12a

Association rate constant for nuclease-active Cas12a is used from Strohkendl et al. (2018). k_{bind} values for nuclease-active Cas12a were observed to be similar for nuclease-inactive Cas12a, suggesting that association rates are not affected by DNA cleavage (Yamano et al., 2016). This justifies our usage of nuclease-active Cas9 binding for the smaller subset assuming association rates are not affected by the cleavage of DNA.

4 Off-target measurements

Alkan et al. (2018) introduced two off-targeting parameters, CRISPRoff and CRISPRspec. CRISPRoff gives the confidence score of the off-target and is approximated using the free energy of RNA-DNA binding. A low CRISPRoff score for a particular target indicates poor binding, whereas a high score indicates stable binding and high chances of cleavage.

$$\text{CRISPRoff}[g, t_{\text{off}}] = -\Delta G_B[g, t_{\text{off}}] \quad (6)$$

CRISPRspec gives the specificity score of the guided RNA which uses energy of all possible off-targets.

$$\text{CRISPRspec}[g, T_g] = -\log_{10} \left(\frac{\sum_{t \in T_g \setminus \{t_{\text{on}}\}} e^{-\beta \Delta G_B[g, t]}}{\sum_{t \in T_g} e^{-\beta \Delta G_B[g, t]}} \right) \quad (7)$$

where numerator accounts for the contributions due to all possible off-targets and denominator accounts for contributions of all potential interactions. A high CRISPRspec score means the guide RNA is more specific.

4.1 Binding stability and off-targeting measurements

CIRCLE-seq database of 18 gRNAs and 7200 off-target sequences were used in Alkan et al.(2018) to compare CRISPRoff and CRISPRspec performances with other off-target scoring methods. The database had information of experimental off-target read counts and CRISPRoff scores are expected to be positively correlated with the read counts. This is because, the more stable an off-targeted binding is, CRISPRoff will be higher and so are the read counts. CRISPRoff outperformed all other methods with a positive Pearson coefficient of 0.43, whereas the second best method had a 0.3 Pearson coefficient.

CRISPRoff scores were measured for the CIRCLE-seq gRNAs with three different cases of binding stability: i) Same as Alkan et al.'s(PAM-proximal region weighed more), ii) Binding stability array in reverse(PAM-distal regions are weighed more)

in comparison to the PAM-proximal regions), iii) Without binding stability(PAM-proximal and PAM-distal nucleotides bind similarly).

CRISPROff scores without binding stability are not informative and poorly correlated ($\rho = 0.15$), while the positive correlation is good for the other two cases.

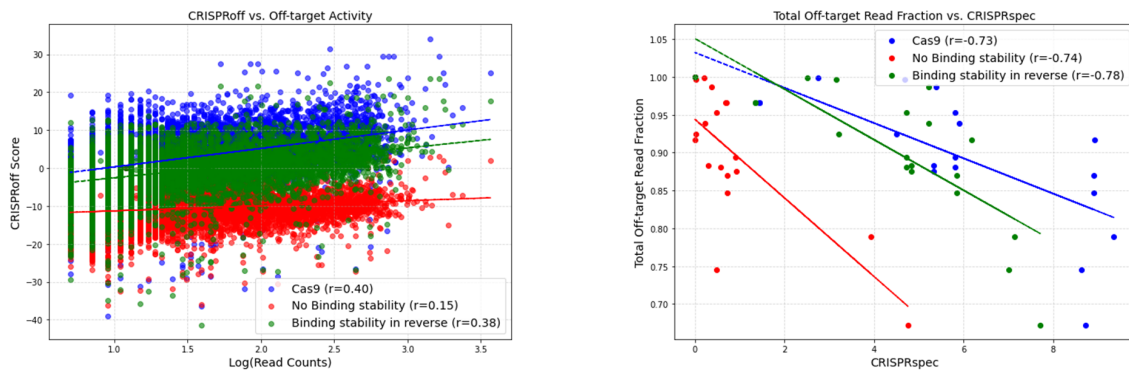


Figure 6: CRISPROff(left) and CRISPRspec(right) for different cases of binding stability in Cas9 vs off-target read counts in CIRCLE-seq gRNAs

Similar to CRISPROff, CRISPRspec scores are in close agreement with experimental off-target read counts with a negative correlation($\rho = -0.72$) and performed much better than all other methods (second best, MIT: $\rho = -0.49$). CRISPRspec scores were obtained for three different cases of binding stability. All three cases showed the negative correlation between CRISPRspec and off-target reads. However, when no binding data is used, CRISPRspec scores for most gRNAs fall in the range of 0 to 1, making it difficult to distinguish highly specific ones. The scores and the correlation are similar for the other two cases. Even though, other two cases show better performance, difference in CRISPRspec between certain gRNAs are huge and can be inaccurate when binding affinities are reversed.

4.2 Cas9 and Cas12a comparison

21 gRNA common for used in both Cas9 and Cas12a systems were used for this purpose. Complementary sequences were taken as the on-target ones while off-targets were generated with upto 4 mismatches. This resulted in a generation of 400,000 off-target sequences to nullify any bias possible. CRISPRspec scores of the first 4 gRNAs suggest that Cas9 is more target specific when compared to Cas12a.

The values observed here are small owing to the use of all possible off-targets most of them which are stable.

Also the CRISProff scores suggest that CRISPR-Cas9 systems are more stable for most compared to the off-targets with CRISPR-Cas12a.

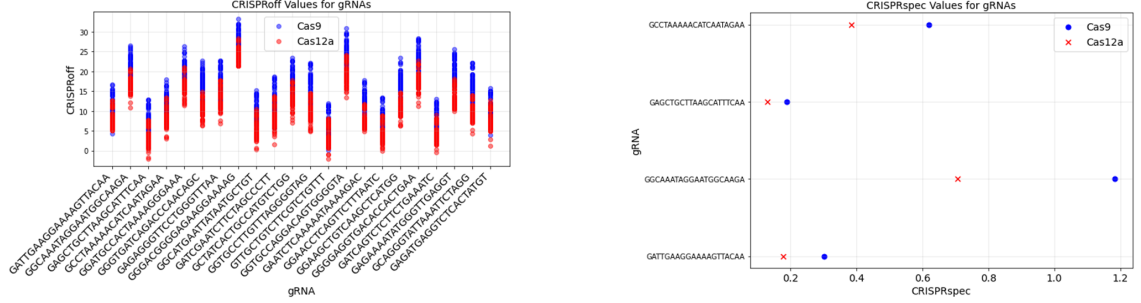


Figure 7: CRISProff(left) and CRISPRspec(right) comparison for Cas9(blue) and Cas12a(red) gRNAs

5 Conclusion

This study made use of the the approximate binding energy model to compare between Cas9 and Cas12a. The key result was that Cas9 is more target specific than Cas12a. This is in contrary to the assumption that Cas12a is better suited for on-target DNA cleavage(Strohkendl et al., 2018). There could be possible errors in the determination of binding stability, partially due to small dataset. Also the association rate for the smaller dataset for Cas9 and Cas12a were obtained using fraction bound of nuclease-active Cas protein, where cleavage of DNA occurs. Kinetics of binding and unbinding mechanism shows that R-loop dissociation is rapid in Cas12a-RNA complex in comparison to the slow unbinding of Cas9-RNA complex. This means that DNA cleavage is more likely when Cas12a is used, thus suggesting it to be more target specific. That being said, Cas12a cleavage is less precise as the cuts are at different positions in both strands of the DNA. Binding energy model stands out when it is used as a standalone for Cas9 systems, however it is to be seen if cleavage rate can be used to compare Cas9 and Cas12.

6 References

1. Nuñez, J. K., Harrington, L. B., Kranzusch, P. J., Engelman, A. N., Doudna, J. A. (2015). Foreign DNA capture during CRISPR–Cas adaptive immunity. *Nature*, 527(7579), 535–538.
2. Koonin, E. V., Makarova, K. S. (2019). Origins and evolution of CRISPR–Cas systems. *Philosophical Transactions of the Royal Society B*, 374(1772), 20180087.
3. Xu, X., Duan, D., Chen, S. J. (2017). CRISPR-Cas9 cleavage efficiency correlates strongly with target-sgRNA folding stability: from physical mechanism to off-target assessment. *Scientific reports*, 7(1), 143..
4. Alkan, F., Wenzel, A., Anthon, C., Havgaard, J. H., Gorodkin, J. (2018). CRISPR-Cas9 off-targeting assessment with nucleic acid duplex energy parameters. *Genome biology*, 19, 1–13.
5. Boyle, E. A., Andreasson, J. O., Chircus, L. M., Sternberg, S. H., Wu, M. J., Guegler, C. K., ... Greenleaf, W. J. (2017). High-throughput biochemical profiling reveals sequence determinants of dCas9 off-target binding and unbinding. *Proceedings of the National Academy of Sciences*, 114(21), 5461–5466.
6. Strohkendl, I., Saifuddin, F. A., Rybarski, J. R., Finkelstein, I. J., Russell, R. (2018). Kinetic basis for DNA target specificity of CRISPR-Cas12a. *Molecular cell*, 71(5), 816–824.
7. Sugimoto N, Nakano S, Katoh M, Matsumura A, Nakamuta H, Ohmichi T, Yoneyama M, Sasaki M. Thermodynamic parameters to predict stability of RNA/DNA hybrid duplexes. *Biochemistry*. 1995;34(35):11211–6.
8. Watkins NE, Kennelly WJ, Tsay MJ, Tuin A, Swenson L, Lee HR, Morosyuk S, Hicks DA, Santalucia J. Thermodynamic contributions of single internal rA·dA, rC·dC, rG·dG and rU·dT mismatches in RNA/DNA duplexes. *Nucleic Acids Res*. 2011;39(5):1894–902.

9. Hofacker, I. L. (2003). Vienna RNA secondary structure server. *Nucleic acids research*, 31(13), 3429-3431.
10. Turner DH, Mathews DH. NNDB: the nearest neighbor parameter database for predicting stability of nucleic acid secondary structure. *Nucleic Acids Res.* 2010;38(Database issue):280–2.27.
11. SantaLucia J, Hicks D. The thermodynamics of DNA structural motifs. *Annu Rev Biophys Biomol Struct.* 2004;33:415–40.